

Energy transfer in chemical reactions

Learning outcomes

By the end of this section you should be able to:

- Describe the law of conservation of energy and its application to chemical reactions.
- Identify system and surroundings and direction of energy transfer for a chemical reaction.
- Explain the difference between heat and temperature.

Fire has been an important tool for human survival and development throughout history and has been an important part of many cultures and religions. Despite its significance, fire was considered quite mysterious throughout much of history. This is because it was always viewed as a physical substance or entity. Instead, fire is better described as an event or process that releases energy in the form of heat, sound and light. It is a chemical reaction that results in this process when a fuel (such as wood) reacts with oxygen in the presence of an ignition source.

A fire triangle, as shown in **Figure 1**, is often used to display the three components necessary for fire. The triangle can be simple to represent a small fire or can be scaled up to consider fire spread over landscapes. It is important for us to consider these larger scale fire triangles as global climates change, influencing more factors involved in the 'wildfire' and 'fire regime' triangles.

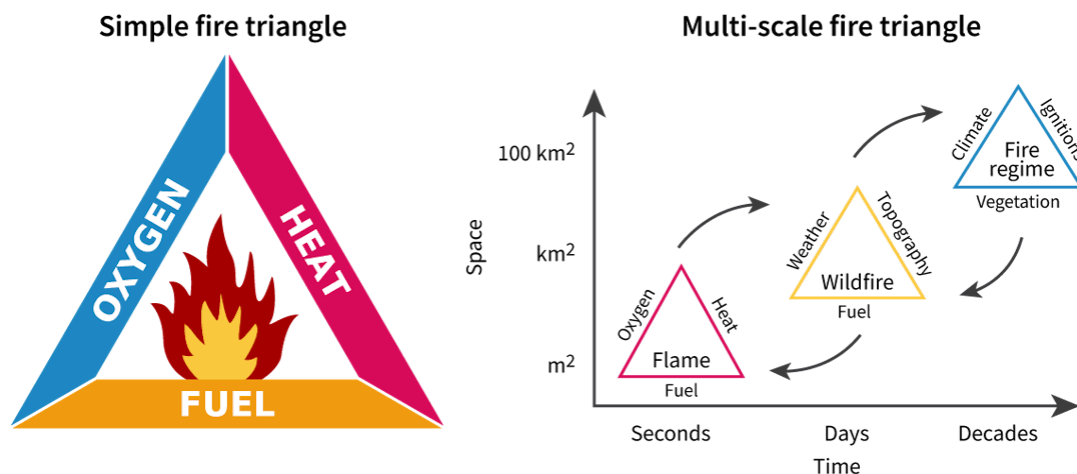


Figure 1. Simple and multi-scale fire triangles.

 More information for figure 1

The image depicts two concepts related to fire dynamics. On the left is a classic fire triangle with three sides labeled 'Oxygen', 'Heat', and 'Fuel', with a flame symbolizing fire in the center. On the right, there is a multi-scale representation of the fire triangle concept, shown as a series of arrows pointing from a smaller red triangle to a medium yellow triangle and finally a larger blue triangle, indicating the progression or scaling of fire over landscapes.

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But how does a chemical reaction from fuel make a fire? How does it produce heat? Why does the temperature of the environment around a fire increase? What is the difference between heat and temperature?

What is energy?

Energy is an important term in chemistry. The term energy used with reference to energy levels that electrons occupy is covered in [section S1.3.1](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>), and ionisation energy considered as a quantitative measurement of the ease to which an electron can be removed from an atom is described in [section S3.1.3b](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/>). But what exactly is energy? How is it measured and what units does it have?

Energy is the ability to do work or cause a change and is measured in units of joules (J). Energy exists in many different forms. **Figure 2** illustrates some different energy forms categorised into energy from movement, stored energy and energy we can experience.

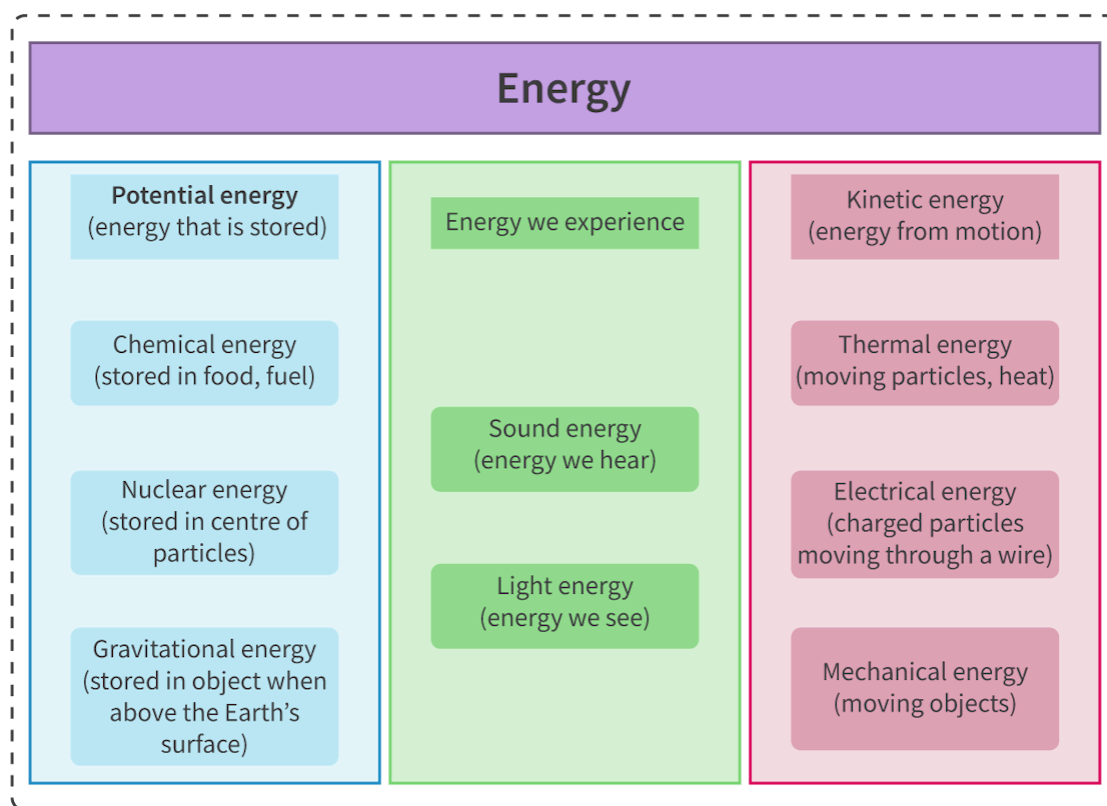


Figure 2. Different forms of energy.

 More information for figure 2

The diagram categorizes different forms of energy. It is divided into three columns.

1. **Potential Energy (energy that is stored):**
2. **Chemical energy:** Stored in food and fuel.
3. **Nuclear energy:** Stored in the center of particles.
4. **Gravitational energy:** Stored in an object when above the Earth's surface.
5. **Energy We Experience:**
6. **Sound energy:** Energy we hear.
7. **Light energy:** Energy we see.
8. **Kinetic Energy (energy from motion):**

9. **Thermal energy:** Moving particles, heat.
10. **Electrical energy:** Charged particles moving through a wire.
11. **Mechanical energy:** Moving objects.

The diagram visually shows the relationship between different energy types, categorizing them into stored, experienced, and motion-related energy.

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Whereas energy is a term most of us are familiar with, it is a strange concept to define on its own. More commonly, we are familiar with the transfer of energy. The three main processes for energy transfer include:

1. **Conduction:** transfer of energy by direct contact
2. **Convection:** transfer of energy by the movement of fluids
3. **Radiation:** transfer of energy by electromagnetic waves.

These three processes for energy transfer are illustrated in **Figure 3** using a pot of water boiling as an example to show how thermal energy can be transferred.

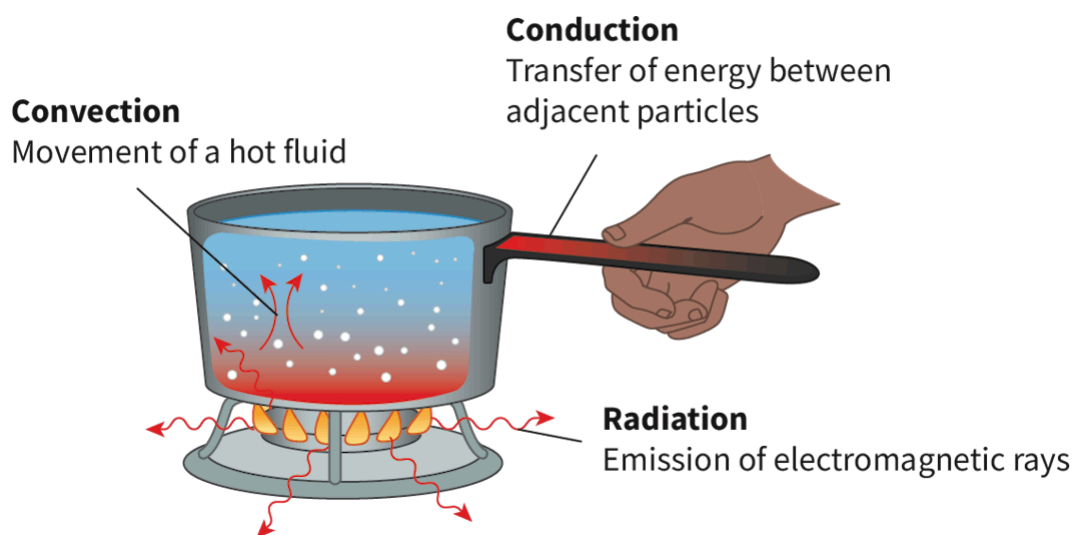


Figure 3. Transfer of thermal energy through conduction, convection and radiation.

 More information for figure 3

The diagram illustrates the transfer of thermal energy using a boiling pot of water. It highlights three methods: conduction, convection, and radiation. Conduction is shown by the hand holding the pot handle, indicating energy transfer between adjacent particles, like from the pot to the hand. Convection is depicted by the movement of hot water within the pot, represented by arrows rising through the water, suggesting the movement of hot fluid upwards. Radiation is illustrated by wavy arrows emanating from the flame below the pot, signifying the emission of electromagnetic rays heating the pot from beneath.

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When a transfer of energy occurs, energy may change from one form to another, but, the total amount of energy will always remain the same in a closed system. This is known as the law of conservation of energy. For example, when we turn on a light bulb, energy is transformed from electrical energy into light, thermal and some sound energy. As a society we place emphasis on the energy transferred to the desired energy form (light) and rate efficiency of materials on that. This rating of efficiency makes it seem like energy is lost, but it is not. It has also been transformed into less useful forms. The total amount of energy remains constant.

Physical changes and energy transfer

Section S1.1.2a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/>) covers physical changes such as melting and freezing. Let's think about what is happening in terms of energy transfers during melting and freezing. When you hold an ice cube in your hand, the solid water is converted to liquid water as it melts. How does this happen? How does your hand feel when this process is happening? Energy is being transferred from your hand to the ice cube to weaken the attraction between particles, allowing the solid to become a liquid. Now, what about the formation of an ice cube? Have you ever noticed that the outside of a freezer is often warm? Can you explain what is happening to make liquid water freeze into ice? Energy is being transferred from the liquid water out of the freezer making it warm to the touch.

These types of energy transfer are not only possible for physical processes but also chemical reactions. As this is the first introduction of chemical reactions so far in the course, let's first look at some of the basics before we look more closely at energy transfers involved in chemical reactions.

Chemical reactions

We interact with all types of substance in our lives, from the fresh air we breathe, to the hot water we wash ourselves with and the soil we walk on. We are constantly interacting with elements, compounds and mixtures each and every day. [Section S1.1.1a \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) shows that compounds consist of atoms of different elements chemically bonded together in a fixed ratio. But how do we form a compound starting with elements? How do we make new compounds from existing compounds?

Elements can undergo a chemical reaction with other elements to form a compound. Compounds can also undergo chemical reactions with other elements or compounds to form new compounds. A chemical reaction is a process where a rearrangement of atoms occurs forming new substance(s) with different properties. The initial substances used in a chemical reaction are called reactants and the new substances formed as a result of the chemical reaction are called products.

We can use a chemical equation to represent a chemical reaction. **Figure 4** shows a chemical equation for the formation of the compound carbon dioxide, CO₂, from its elements carbon, C, and oxygen, O₂.

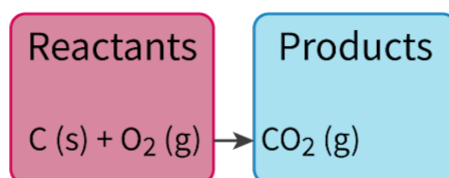


Figure 4. Chemical equation for the reaction forming carbon dioxide.

 More information for figure 4

The image is a diagram depicting a chemical equation for the formation of carbon dioxide (CO₂). On the left, a red box labeled "Reactants" contains the elements C (s) plus O₂ (g). An arrow points from this box to a blue box on the right labeled "Products," which contains CO₂ (g). The diagram illustrates the chemical reaction where solid carbon (C) reacts with gaseous oxygen (O₂) to form gaseous carbon dioxide (CO₂).

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Transfer of energy in chemical reactions

Think about chemical reactions you may have encountered such as wood burning in a campfire, plants photosynthesising, metabolising food or cooking an egg. What do all of these chemical reactions have in common? They all involve a transfer of energy. **All** chemical reactions involve a transfer of energy between the system and the surroundings, while total energy is conserved. In a chemical reaction, energy cannot be created or destroyed; instead, it is either converted from one form to another or transferred between the system and its surroundings. The system refers to the components involved in the chemical reaction and the surroundings encompasses everything outside the system. **Figure 5** shows the system and surroundings for the processes mentioned earlier.

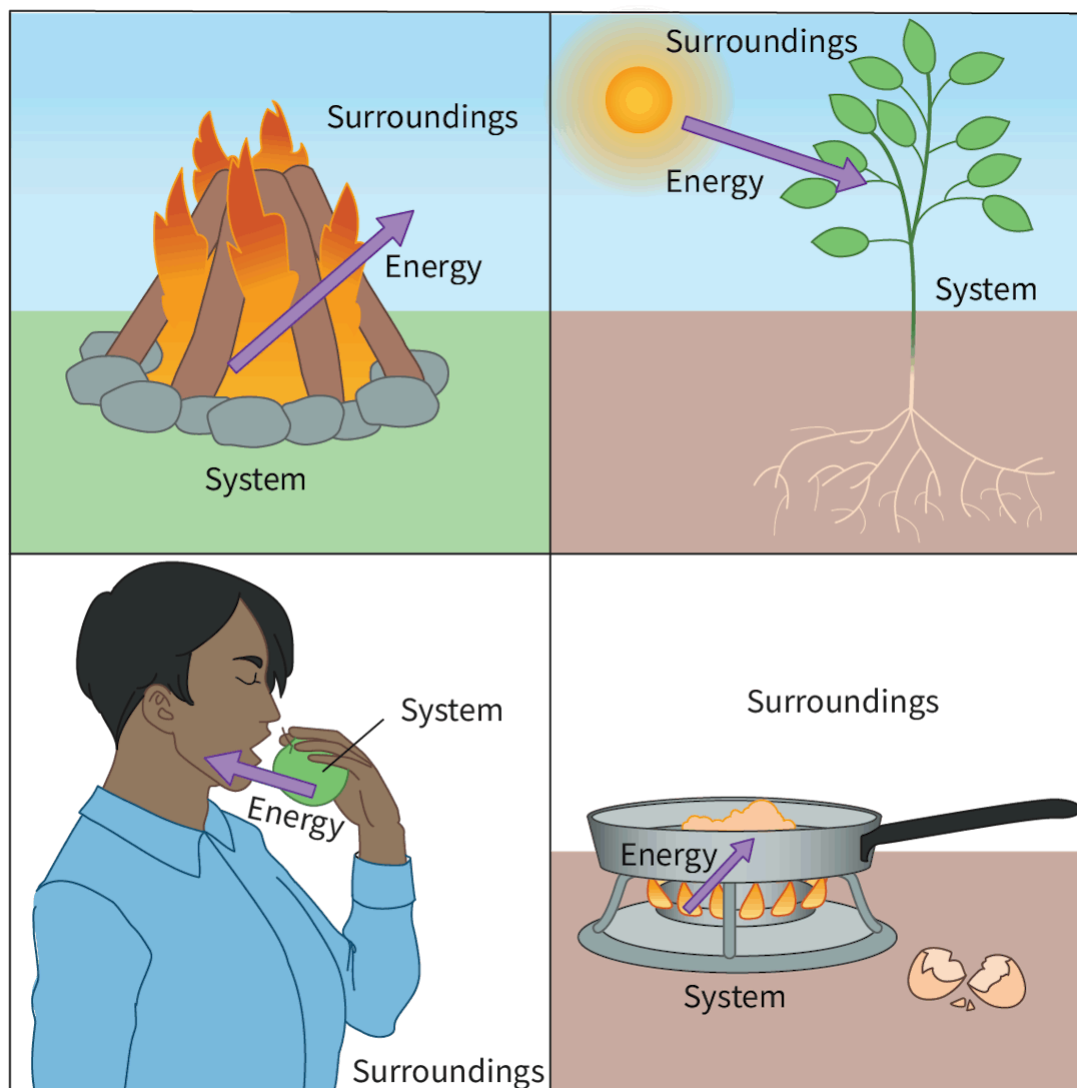


Figure 5. Common everyday chemical processes that involve a transfer of energy between system and surroundings.

 More information for figure 5

This illustration shows four common chemical processes where energy transfer occurs between a system and its surroundings. The image is divided into four quadrants:

1. Top Left: A campfire with logs is depicted, flames rising into the air. An arrow labeled "Energy" points away from the "System" (the fire and logs) towards "Surroundings."
2. Top Right: A plant rooted in the ground under the sun is illustrated. An arrow labeled "Energy" points from the "Surroundings" (the sun) towards the "System" (the plant).
3. Bottom Left: A person is depicted eating, with an arrow labeled "Energy" pointing from a loaf of bread ("System") into their mouth, with "Surroundings" indicated around the person.
4. Bottom Right: A frying pan on a stove with a flame underneath is shown. An arrow labeled "Energy" points from the "Surroundings" (the heat source) into the "System" (the pan). This quadrant also shows broken eggshells, implying that something is being cooked in the pan.

This diagram illustrates how energy is transferred in different processes: combustion, photosynthesis, digestion, and cooking.

[Generated by AI]

In a reaction, such as the combustion reaction of methane (sometimes referred to as natural gas), methane reacts with oxygen to produce carbon dioxide and water via the following chemical reaction.



The chemical potential energy stored in the bonds of the reactants is transformed into thermal energy and light energy during the combustion process. The total amount of energy is conserved.

Nature of Science

Aspect: Patterns and trends

Here we have looked at how energy is able to change forms but the total amount of energy will be constant. This means that in a chemical reaction, there will be the same amount of total energy before and after a chemical reaction. This principle is so fundamental and important in science that it is called a law. This is the **law of conservation of energy** and is also known as the **first law of thermodynamics**.

If you study DP physics, you will discover that this law does not allow us to explain nuclear reactions, where there is an unexpected gain of total energy when some mass is lost. Albert Einstein proposed that what is happening is that the lost mass is instead converted into energy. This shows that even laws that appropriately describe much of our universe themselves have their limitations.

During a chemical reaction, energy may be released or absorbed in various forms such as heat, light or electrical energy. You have probably encountered each of these forms in different chemical reactions from processes such as combustion of wood in a campfire (heat), luciferin reacting in fireflies (light) and the oxidation and reduction reactions in a battery (electrical). These are shown in **Figures 6 to 8**.



Figure 6. Processes that release energy: fire.

Credit: Teemu Tretjakov, Getty Images



Figure 7. Processes that release energy: fireflies.

Credit: Ali Majdfar, Getty Images

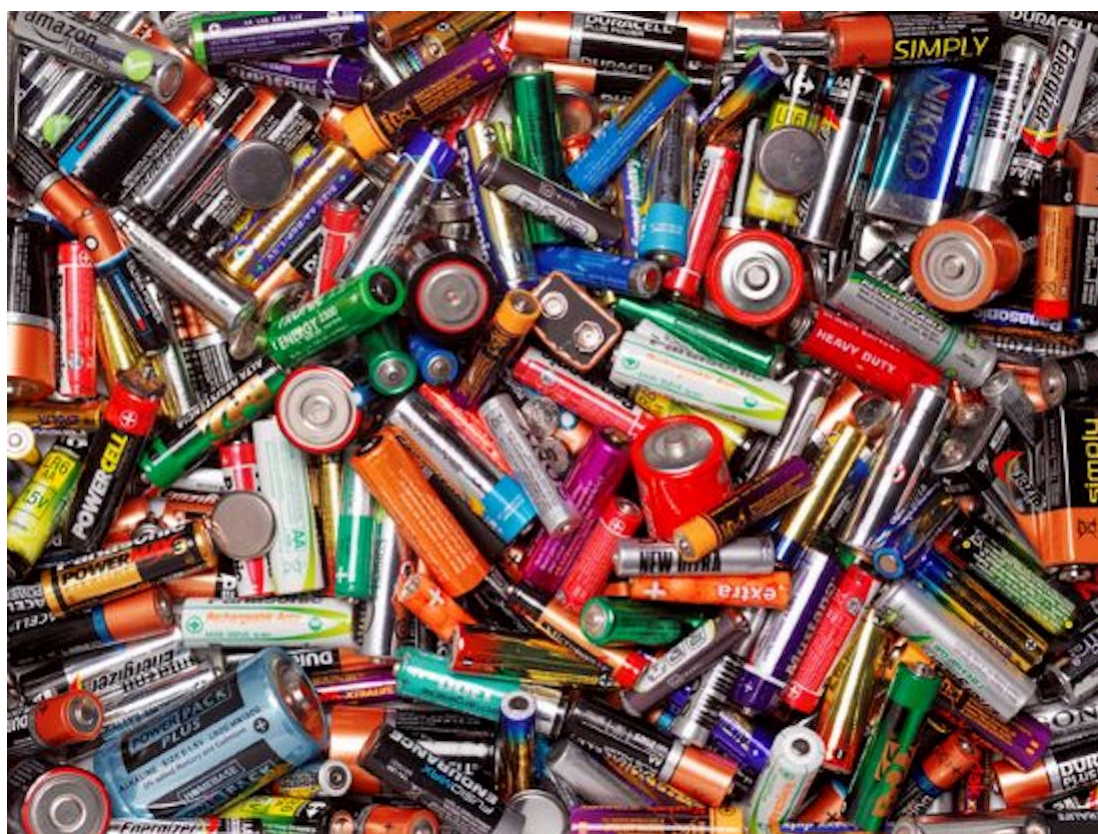


Figure 8. Processes that release energy: batteries.

Credit: Peter Dazeley, Getty Images

Heat and temperature

Consider a chemical reaction taking place in a glass beaker that releases energy in the form of heat. What are some observations you would be able to make? How would energy be measured? Think about the equipment you have in your classroom laboratory: is there an 'energy meter'? If heat is being released by a reaction in the beaker and feels warm, what simple tool can you use to record quantitative data? When you have a fever, what instrument do you use to confirm that your body's temperature has increased as your body uses additional energy to fight an illness?

Temperature is a quantitative measurement often associated with heat. Section S1.1.3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/temperature-and-kinetic-energy-id-43068/>) mentions that temperature is a measure of the average kinetic energy of particles measured in units of K or °C. Heat is the energy transferred from a region of high temperature to a region of low temperature measured in units of J. These two terms are related but are different as illustrated in **Figure 9**. Temperature alone does not determine the amount of heat transfer. We will look at the other variables that are involved in heat transfer in section R1.1.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpy-and-heat-transfers-id-43576/>).

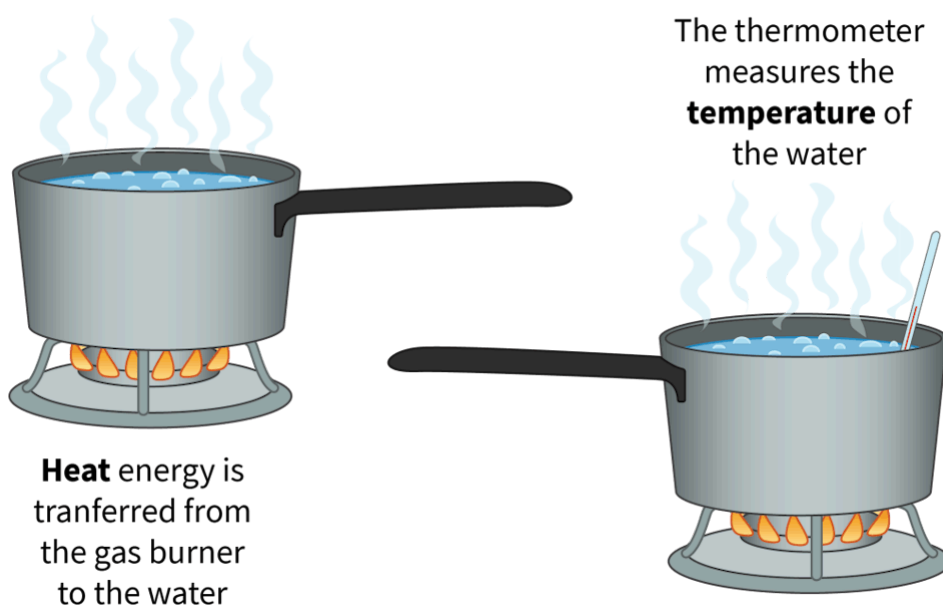


Figure 9. Comparison between heat and temperature.

 More information for figure 9

The image depicts two scenarios involving pots on stoves to illustrate the difference between heat and temperature. On the left, a pot is placed on a gas burner with flames, indicating heat transfer from the burner to the water in the pot. Above this pot, steam is rising, and the text reads, "Heat energy is transferred from the gas burner to the water." On the right, another similar pot is on a gas burner with steam rising, but this time, a thermometer is placed in the water. The text next to this pot states, "The thermometer

measures the temperature of the water." The image aims to show that while heat energy is transferred to increase the water's temperature, the thermometer measures the current temperature of the water.

[Generated by AI]

Theory of Knowledge

You have been provided with a lot of examples involving fire: an extreme release of heat that occurs in certain chemical reactions. The opposite of heat would be coldness, but what exactly is 'coldness'? Is it only the absence of heat, or is there something more?

There is historical medical evidence from many ancient cultures that viewed various maladies, such as cancer and infection, as a build-up of 'heat' in the body with the remedy to provide coldness in the form of cold water, ice baths or certain herbs designated as cold herbs, such as peppermint. Similarly, there were other illnesses that were thought to be caused by a build-up of coldness in the body, such as coughs: even the common cold! The treatment for these 'colds' was to provide heat such as warm clothing and herbs such as cinnamon that were thought to bring heat into the body.

Did isolated Indigenous societies intrinsically understand complex scientific processes, such as energy transfer?

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Thinking skills — Providing a reasoned argument to support conclusions
- **Time required to complete activity:** 30—45 minutes
- **Activity type:** Pair activity

Heat transfer and energy conservation

Aim: To investigate the difference between temperature and heat, the difference between the system and surroundings and the law of conservation of energy

To perform this simple investigation, you will need the following materials:

- Two identical metal cans

- Two thermometers
- Hot water
- Cold water
- Ice
- Stopwatch

Procedure:

- Fill each can up halfway with water: one can with hot water, the other with cold water.
- Measure and record the initial temperatures of water in each can.
- Place an ice cube into each can and start your stopwatch, stirring gently with the thermometer.
- Measure and record the temperatures of water in each can every minute for 10 minutes, or until the ice cubes melt completely.

Results and discussion:

- Represent your data in a well-organised data table.
- Calculate the change in temperature of the water for each can.
- Compare the changes in temperature for both cans and their water contents. Discuss which one had a greater change in temperature including an explanation of why.
- Explain what happened to the heat energy in each can during the experiment. Make sure you mention which part of the system gained heat and which part lost heat. Provide evidence for your reasoning.
- State the law of conservation of energy and discuss how it applies to this experiment.